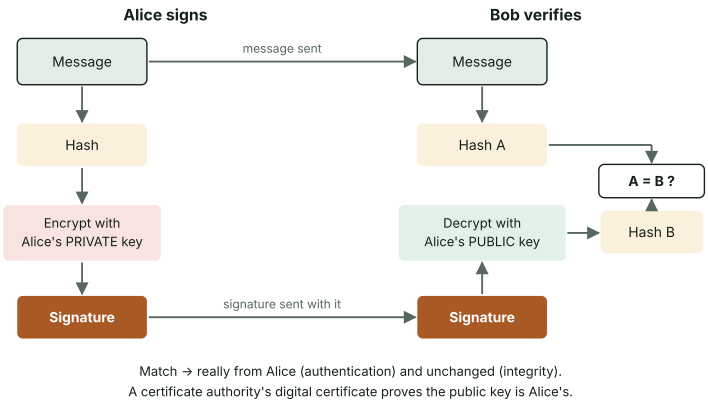
Network security: threats, firewalls, encryption and authentication

IS13 FACT FILE

Specification: students should be able to

- Describe threats to networks and data
- Explain the use of firewalls and proxy servers
- Explain symmetric and asymmetric encryption, digital signatures and digital certificates
- Describe authentication methods and the use of virtual private networks (VPNs)

Key words: malware, phishing, denial of service, firewall, proxy server, symmetric encryption, asymmetric encryption, public key, private key, digital signature, digital certificate, authentication, VPN



Creating a digital signature by hashing a message and encrypting the hash with the sender's private key, and verifying it by decrypting with the sender's public key and comparing hashes

Threats and defences

Networks face malware, phishing, hacking, denial-of-service attacks, interception of data and insider misuse. Defence in depth combines technical measures – firewalls, anti-malware, encryption, patching, authentication, VPNs – with procedures such as staff training, access rights, backups and acceptable use policies. A firewall filters traffic by rules about addresses, ports and protocols. A proxy server makes web requests on behalf of users, hiding internal addresses, filtering content and caching pages.

Encryption

Symmetric encryption uses one shared secret key for both encryption and decryption; it is fast but the key must be distributed securely. Asymmetric encryption uses a public key, which anyone can have, and a private key, kept secret by its owner; data encrypted with one can only be decrypted with the other. Because asymmetric encryption is slow, protocols such as TLS use it to exchange a symmetric session key and then encrypt the session symmetrically.

Signatures and certificates

A digital signature is created by hashing a message and encrypting the hash with the sender's private key. The receiver decrypts it with the sender's public key and compares it to a fresh hash; a match proves the sender's identity and that the message was not altered. A digital certificate issued by a certificate authority binds an organisation's identity to its public key, so users can trust that the key genuinely belongs to that organisation.

T Key terms from the mark schemes

Firewall
Hardware or software that monitors network traffic and allows or blocks it according to a set of security rules.

Proxy server
A server that acts as an intermediary between users and the internet, hiding internal addresses, filtering content and caching pages.

Symmetric encryption
Encryption in which the same secret key is used both to encrypt and to decrypt data.

Asymmetric encryption
Encryption using a mathematically related key pair: data encrypted with the public key can only be decrypted with the private key, and vice versa.

Digital signature
A hash of a message encrypted with the sender's private key, used to verify the sender's identity and that the message has not been altered.

Digital certificate
An electronic document issued by a certificate authority that binds a public key to the identity of its owner.

Virtual private network (VPN)
An encrypted connection across a public network that lets a remote user access a private network securely.

Exam focus

Be precise about keys: confidentiality uses the recipient's public key; signing uses the sender's private key. Many answers lose marks by saying 'the key' without saying whose and which. When describing a firewall, say what it checks and what it does (blocks or allows according to rules). For 'evaluate' or 'recommend' questions, match each measure to a specific threat in the scenario and mention a limitation – no single measure is sufficient.

? Check yourself

1. Describe the purpose of a firewall. [2]
2. Give two functions of a proxy server. [2]
3. Explain the key distribution problem in symmetric encryption and how asymmetric encryption overcomes it. [3]

Answers: 1. A firewall monitors incoming and outgoing network traffic (1) / and allows or blocks it according to a set of rules, e.g. based on IP addresses, ports or protocols, to prevent unauthorised access (1). 2. Any two from: makes requests to the internet on behalf of users, hiding internal IP addresses (1) / filters web content / blocks access to unsuitable sites (1) / caches popular web pages to speed up access and save bandwidth (1) / logs users' web activity (1). 3. In symmetric encryption the same secret key is used to encrypt and decrypt, so the key must somehow be shared with the receiver (1) / if it is sent over the network it could be intercepted, and anyone with it can read the messages (1) / asymmetric encryption uses a key pair: the public key can be published openly and used to encrypt, but only the matching private key, which is never shared, can decrypt (1). Accept: asymmetric encryption is used to exchange a symmetric session key securely (as in TLS).